How a single genome can tell *which kind* of first-degree incest produced a girl — step by step

A plain-language companion to the methods note. Robust but readable: every step is something the simulation in this repository actually shows. "First-degree" means parents as closely related as possible short of being the same person: father–daughter (FD), mother–son (MS), or brother–sister (SS).

Step 0 — The puzzle

When a child is conceived by close relatives, large stretches of their two genome copies are identical, because both copies descend from the same recent ancestor. These stretches are **runs of homozygosity (ROH)**. Clinical genetic tests see them and can say *"the parents are first-degree relatives."*

But here is the catch. The standard summary number — the fraction of the genome that is autozygous, called **F** — is **¼ for all three** first-degree unions. Father–daughter, mother–son, brother–sister: identical on that number. So the usual reading of the test **cannot tell which kind of union it was.** That distinction matters enormously: a father–daughter case means an adult abusing a child; a sibling case is a different situation legally, clinically, and ethically.

The question of this work: **is the missing information really gone, or is it hiding somewhere the standard summary throws away?** It's hiding — in two places.

Step 1 — Why the autosomes seem blind

The 22 non-sex chromosomes (autosomes) are inherited symmetrically: you get one copy from each parent, and both parents shuffle (recombine) their two copies before passing one on. For all three first-degree unions, the total amount of autozygous genome works out to the same ¼. That's why the headline F number is uninformative. Hold that thought — the autosomes are *not* actually blind, but you have to look past the total (Step 6).

Step 2 — The X chromosome breaks the symmetry

The X is inherited in a lopsided way:

- A **father gives his single X to his daughters completely intact** — no shuffling. (He has only one X; he hands it over whole.)
- A **son gets no X at all from his father** (he gets the Y instead).
- A **mother shuffles her two X's** and passes one mixed copy to each child.

Because of this, in a **girl**, how much of her two X's match depends on **which sexes lie along the path connecting her parents back to their shared ancestor** — not just how close that ancestor is. We call that path the **sex-path**.

Step 3 — The three unions give three different X signatures

Run the inheritance forward for a daughter:

- **Father–daughter (FD):** her father hands her his X *intact*. That same X also sits, intact, inside her mother (because the mother is the father's daughter). So one of the girl's X's is an unshuffled copy that the other X is trying to match. Result: on average **half** the X is autozygous (**F_X = ½**), in one or a few long blocks.
- **Mother–son (MS):** both of the girl's X's are shuffled copies of the *same* grandmother-mother's two X's — but shuffled in two separate events. They still match **half** the time on average (**F_X = ½**), but broken into *more* pieces, because two shufflings make more seams.
- **Brother–sister (SS):** the matching can only happen through the shared grandmother's X, and the grandfather's X (which rides along on one side) can never match. Result: only **a quarter** of the X is autozygous (**F_X = ¼**).

So just from the *level* of X matching: **sib–sib (¼) stands apart from the two parent–child types (½).** That's the first real signal the standard F number hid.

Step 4 — Reading the pattern as "seams"

Picture the girl's X as a strip that is either "matching" or "not matching" as you walk along it. The points where it flips are **seams**, and each seam is a recombination (shuffling) event. The three stories predict different numbers of seams:

- **FD:** seams come from **one** shuffling → few seams.
- **MS:** seams come from **two** shufflings → about twice as many seams.
- **SS:** fewer matching regions overall, with built-in non-matching blocks.

The recombination **map** tells us exactly how likely a seam is at each spot on the X. So instead of guessing, we can **calculate the probability of the observed seam-pattern under each story** and compare. That probability is the *likelihood*, and the ratio of two likelihoods is the weight of evidence for one story over another. No guessing, no simulation — a formula read off the map.

Step 5 — Why father–daughter vs mother–son is genuinely stuck

When you write down that calculation for FD vs MS, something clean happens: the **map cancels out**. The only thing that ends up mattering is the **number of seams** — FD averages about 1.8, MS about 3.5. From a single genome you see *one* number of seams, and "about 1.8" versus "about 3.5" overlap a lot. So FD vs MS is **near a coin flip (~0.71 at best)** — and we can prove it's not a tooling limitation;

it's the information limit. The fine detail of *where* the seams fall, which helps elsewhere, gives nothing here.

For **FD vs SS** (and MS vs SS) the map does **not** cancel — the stories differ in the total amount of matching and in their structure — so the full pattern helps. This is the good news, because…

A closer look at the seam count: the "zero-shuffle" corner

Every seam is a shuffling (recombination) event, and on the female X the number of those events per copy is a lottery: about **17% of the time zero, 30% once, 27% twice, 16% three times**, and so on (it averages 1.76). The whole story turns on the **zero** outcome — and each union reaches it differently:

- **Father–daughter** depends on **one** such lottery (the mother's single transmission). When it comes up zero, the mother hands over one of her two X's *whole* — a coin flip between her father's copy (→ the girl's X is **entirely matching**) and her mother's copy (→ **entirely non-matching**). Each happens **8.6%** of the time.
- **Mother–son** depends on **two** lotteries both coming up zero — much rarer (**1.5%**). That's the simple reason a fully-homozygous X points to father–daughter over mother–son by about **6 to 1**.
- **Brother–sister** is lopsided: the grandfather's X rides along on one side and can *never* match, so an entirely-matching X is essentially impossible (**0.13%**), while an entirely-*non*-matching one is common (**18%**).

So the two extremes point opposite ways: an **entirely homozygous** X shouts *father–daughter* (about **68:12:1** over mother–son and sibling), while an **entirely outbred-looking** X leans *sibling* (about **12:6:1**). These aren't simulation guesses — they are exact formulas from the recombination process, confirmed by the simulation to the third decimal. And note the average is unmoved — it stays ½, ½, ¼ no matter how the shuffling lottery falls — which is exactly why the plain average hides the answer and the *spread* reveals it.

Step 6 — The autosomes are not blind after all (two hidden signals)

Go back to the autosomes. The *total* is ¼ for everyone, but the **sizes and number of the ROH pieces differ**, for two reasons:

(a) Depth. An autozygous piece gets chopped up by every shuffling between the child and the shared ancestor. Parent–child loops pass through **3** shufflings; sibling loops pass through **4** (because siblings share *two* grandparents, one extra step on each side). More shufflings → **shorter, more numerous** pieces. So:

union	typical # of ROH pieces	typical size
father–daughter	~26	larger
mother–son	~30	medium
brother–sister	~35	smaller

All add up to ¼ — but the **count** cleanly separates sib–sib from parent–child. And because this is averaged over 22 chromosomes, it's a **reliable** signal, unlike the single, noisy X.

(b) Meiosis sex (the subtle one). Female shuffling produces ~1.6× more seams than male shuffling. A father–daughter loop runs mostly through *male* steps; a mother–son loop runs mostly through *female* steps. So mother–son fragments a bit more than father–daughter even though both have the same depth — giving the autosomes a faint echo of the *same* sex-path story the X tells loudly. It's weak, but real (we predicted a 16% effect and measured 14%).

Step 7 — Putting both chromosomes together: the scoreboard

Combine the autosomes (depth) and the X (sex-path), and ask how well a single genome can tell each pair apart:

question	answer from one genome
father–daughter vs sibling	**~0.87** — well resolved
mother–son vs sibling	~0.77
father–daughter vs mother–son	~0.75 — the stubborn pair

The headline: the **most consequential question — father–daughter versus sibling — is the best-answered one.** The autosomes nail the depth; the X confirms the sex-path. The only pair that stays hard is father–daughter vs mother–son, because they are identical in depth *and* in X level, differing only in the noisy seam-count.

Step 8 — How to crack the last hard pair

Father–daughter vs mother–son is stuck **from the child alone**. But add **one genotyped relative — the birth mother — and it jumps to ~0.91.** The reason is elegant: under father–daughter, the girl's intact paternal X is a perfect, unshuffled copy of one of her mother's X's; under mother–son, it's a *shuffled mix* of both. Counting how many times the girl's paternal X "switches" between her mother's two X's gives the answer: about zero (father–daughter) versus several (mother–son). The child's genome can't show this; the child plus mother can.

And here is why that is so often *practical* rather than theoretical: **the mother is almost always known.** She was pregnant and gave birth, so her identity is rarely in doubt and she is usually available — whereas the **father is frequently unknown or absent** (he is the perpetrator in a father–daughter case, the son in a mother–son case). The lucky part is that this lines up perfectly with the genetics: the one comparison the child's genome can't make on its own — father–daughter versus mother–son — is exactly the one that the *mother's* genome resolves. So the relative you can almost always get is precisely the one you need, and the relative who is usually missing (the father) is one the method doesn't require. (The mother's age is a weaker, secondary hint in the same direction — a mother–son mother is necessarily older — but it's the *availability of her genome*, not her age, that does the real work, and case facts like these should be used as honest up-front priors, never counted twice.)

Step 9 — Why this matters, and the irony

- **Clinically and forensically**, separating father–daughter from sibling is the question that changes what happens next — and it's the one a single genome answers best.
- **The irony:** the X chromosome carries the sex-path signal, yet clinical labs routinely **exclude the X** from homozygosity calculations. One published case even describes a girl from a first-degree union whose X was *entirely* homozygous — exactly the father–daughter fingerprint — noted only in passing. The discriminating chromosome is the one usually set aside.

Step 10 — How solid is this?

- The core inheritance numbers (½, ½, ¼) **match theory exactly** in simulation — that agreement is the built-in correctness check.
- Everything was checked on **two independent recombination maps** (Bhérer and deCODE), which agree to within half a percent — so the conclusions don't depend on map choice.
- The ~0.87 "father–daughter vs sibling" headline is now confirmed by a **single combined simulation** that drops all 22 autosomes *and* the X through each family and lets one classifier read the whole genome — not just by adding the two chromosomes' scores together. And it is a **conservative** number: when we use a more realistic recombination model (where shuffling events repel each other, as they really do), the resolution gets slightly *better*, not worse.
- The honest limits: these are single-genome accuracies; the genotype data here are realistically simulated, not yet real patient data; and turning "first-degree" into a *named* relationship raises real consent and reporting questions that any use must respect.

In one sentence: the autosomes tell you *how deep* the family loop is, the X tells you *which sexes* it ran through, and together — from a single girl's genome — they reveal not just *that* the parents were first-degree relatives, but *which kind*, with the most important distinction being the clearest.